

Multiple Choice Quiz

Comprehensive: From Location-Based to Attribute-Based Discovery

Total Questions: 10

Time Allowed: 20 minutes

Instructions: Choose the **best** answer for each question. Each question carries equal marks.

Question 1

What is the fundamental shift in distributed discovery architecture described in the sources?

- A. From attribute-based discovery to purely location-based addressing
 - B. From a “location-based” model (“Where is it?”) to an “attribute-based” model (“What is it?”)**
 - C. From decentralized systems to fully centralized global directories
 - D. From human-readable names to purely cryptographic identifiers
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Question 2

Why are legacy location-based systems considered insufficient for modern content discovery?

- A. They are too slow even when the exact address is known
 - B. They require too much manual configuration by administrators
 - C. They cannot scale beyond a few thousand users
 - D. They can locate known addresses but cannot discover content based on its characteristics or identity (“I Know It When I See It” problem)**
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Question 3

Describe the “human bottleneck” associated with manual metadata tagging.

- A. Subjective and inconsistent tagging by different people creates “metadata entropy,” causing discovery to degrade into linguistic chance**
- B. It refers only to the high cost of hiring professional metadata specialists
- C. It occurs when machines automatically generate perfect metadata without human input
- D. It is solved by forcing all users to use a single rigid taxonomy

Question 4

In terms of performance and search cost, how does LDAP differ from DNS?

- A. LDAP is always faster and lighter than DNS because it uses attributes
 - B. Both systems have identical search costs and architectures
 - C. DNS uses lightweight, direct lookups on single endpoints, while LDAP requires heavy exhaustive scans of Directory Information Trees and distributed coordination
 - D. LDAP has lower search costs because it avoids any form of hierarchy
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Question 5

Explain the “Pheriby Smith” Paradox and its computational impact.

- A. It describes a situation where all attributes are equally common, causing no performance issues
 - B. It occurs when separating attributes across servers forces the intersection of a unique key with a ubiquitous key, flooding the client with millions of irrelevant matches
 - C. It only affects centralized systems and has no impact on distributed architectures
 - D. It refers to the difficulty of remembering long cryptographic names
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Question 6

What is the primary goal and benefit of “Flattening the Universe” in attribute space?

- A. To increase the number of dimensions to support more complex queries
 - B. To make all data globally visible without any partitioning or indexing
 - C. To eliminate the need for any form of hashing or distributed storage
 - D. To collapse multi-dimensional attribute space into a single dimension, enabling standard hashing while preserving range query capability
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Question 7

What is the “crucial property” of Hilbert Curves as defined in the source context?

- A. They preserve locality: points close together in multi-dimensional space remain close together when mapped to a 1D line
 - B. They randomly distribute data points to achieve perfect load balancing
 - C. They only work effectively in two-dimensional spaces
 - D. They eliminate the need for any form of caching in distributed systems
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Question 8

How does the HyperDex system implement implementation at scale?

- A. It stores all data in a single centralized database for simplicity
 - B. It uses only recursive resolution without any form of hashing
 - C. It maps N-dimensional attributes to a 1D line using space-filling curves and distributes them across a Chord ring, allowing efficient multi-attribute queries without sifting through irrelevant records
 - D. It avoids all forms of indexing to reduce computational overhead
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Question 9

Define the structure of a Resource Description Framework (RDF) triplet and its purpose.

- A. It consists of three identical copies of the same data for redundancy
 - B. It consists of Subject → Predicate → Object, creating a self-describing “Web of Data” where attributes become first-class citizens with structural meaning
 - C. It is used exclusively for encrypting network traffic
 - D. It replaces all forms of hierarchical naming with flat cryptographic identifiers
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Question 10

What is the ultimate architectural trade-off identified in the transition to attribute-based discovery?

- A. Trading human usability for simpler machine addressing and lower computational cost
 - B. Moving from complex mathematical systems back to simple file paths
 - C. Eliminating all forms of metadata to reduce system complexity
 - D. Prioritizing human-centric discovery and usability at the cost of significantly higher global mathematical and computational complexity
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Answer Key

Question	Correct Answer	Explanation
1	B	The fundamental shift is from asking “ Where is it? ” (location) to “ What is it? ” (attributes and meaning).
2	D	Legacy systems (URLs/paths) work

Question	Correct Answer	Explanation
		when you already know the address but fail at discovery based on content characteristics.
3	A	Manual tagging leads to inconsistent metadata ("metadata entropy"), making search unreliable.
4	C	DNS is lightweight and direct. LDAP requires expensive exhaustive scans and heavy coordination.
5	B	The paradox creates massive client-side processing when intersecting unique and common attributes across servers.
6	D	Flattening collapses N-dimensional space into 1D to enable hashing while keeping range queries possible.
7	A	Hilbert curves preserve locality — closeness in multi-dimensional space is maintained on the 1D curve.
8	C	HyperDex uses space-filling curves + Chord ring hashing for efficient multi-attribute queries at scale.
9	B	RDF uses Subject → Predicate → Object triplets to create structural, self-describing meaning.

Question	Correct Answer	Explanation
10	D	The trade-off is gaining powerful human-centric discovery at the cost of high mathematical/computational complexity.

End of Quiz

Correct Answer Distribution: $A = 2$ | $B = 3$ | $C = 2$ | $D = 3$

This comprehensive quiz integrates concepts from location vs. attribute-based discovery, metadata challenges, bottlenecks, Hilbert curves, HyperDex, and RDF.